

Histology of LYMPHOID ORGANS AND IMMUNE SYSTEM

INTRODUCTION

- The human body tissues are subject to constant **threat of invasion of potentially harmful agents and pathogens** such as bacteria, viruses, fungi, protozoa and multicellular parasites via **breaches in the skin or linings of gut, respiratory and urogenital tracts.**
- Consequently, three main lines of defence have evolved:
 1. *protective surface mechanisms*
 2. *non-specific tissue defenses*
 3. *specific immune responses*
 - *mounted by specific immune cells which comprise one of the several body systems (immune system)*

1. PROTECTIVE SURFACE MECHANISMS

- In humans, these provide first line of defence & include:
 - i. **The skin:** that constitutes a relatively impermeable barrier to most micro-organisms unless breached by injury such as abrasion or burning.
 - ii. **The sero-mucous** surfaces of the body such as the conjunctiva and oral cavity, are protected by a variety of antibacterial substances including the enzyme lysozyme, secreted in tears and saliva.
 - iii. The respiratory tract is protected by a layer of **mucus film/surface** which is continuously removed by ciliary action and replaced by goblet cells.
 - iv. **Acidic environment** in the stomach, vagina and to a lesser extent the skin, inhibits the growth of pathogens in these sites.

2. NON-SPECIFIC TISSUE DEFENSES

- activated if the first line of defence does not work.
- damages to tissues usually elicit non-specific tissue responses collectively called **inflammation.**

Inflammation

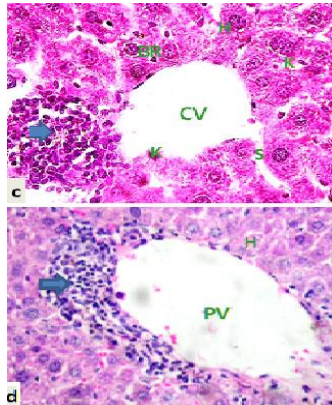
- aimed at removal of any dead tissue and foreign matter and replacement of lost tissue by scar and in some cases regeneration of normal tissue.
- may be acute or chronic

Acute inflammation

- Characterized by vascular changes including capillary dilatation, enhanced permeability and increased blood flow resulting in a production of a fibrin-rich inflammatory exudate.
- Neutrophils, and later macrophages, migrate into the damaged tissues and exert their phagocytic activity to remove tissue debris, foreign material and pathogens.

Photomicrographs of H and E stained liver sections showing chronic inflammatory cellular infiltrations (arrows) around central vein (cv) (c) and portal vein (pv) (d).

Examples of Inflammation/ non-specific tissue defense



From Tilahun et al., 2017, BMC

IMMUNE SYSTEM

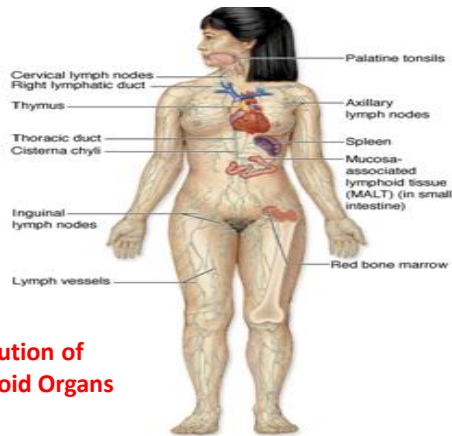
- comprises all the specific and complex immune responses of the body and constitutes one of the **11 organ systems**.
- the ability to distinguish "**self**" (the organism's own molecules) from "**nonself**" (foreign substances) molecules.
- aimed to **neutralize** or **inactivate** foreign molecules (such as soluble molecules as well as molecules present in viruses, bacteria and parasites).
- destroys microorganisms or other cells (such as virus-infected cells, cells of transplanted organs, and cancer cells)
- rarely, reacts against once own normal body tissues or

Cells of the Immune System include:

- Lymphocytes (T, B and NK cells),
- Plasma cells,
- Neutrophils,
- Eosinophils, and cells of the mononuclear phagocytic system (monocytes-macrophages system)
- Antigen-presenting cells (APCs)= a group of very diverse cell types which assist other immune cells
 - APCs include, among other cells, macrophages, Langerhan's cells, lymphoid dendritic cells, Microfold (M) cells, follicular dendritic cells (FDC), B cells, mast cells and granulocytes.

THE IMMUNE CELLS

- "the police men or the solders of one's own body"
- are diffusely distributed throughout the body in the **blood, lymph, epithelial** and **connective tissues**; or
- arranged in small spherical masses called **lymphatic nodules (lymph follicles)** in connective tissues and inside several organs; or
- organized as discrete organs called **lymphoid organs**—the lymph nodes, the spleen, thymus, and the red bone marrow.
- also exists as **lymphoid tissue masses** in the mucosa of the digestive system forming **GALT (tonsils, Peyer's patches, and appendix)**, the respiratory system as **BALT**, and in the urogenital ducts as **UGALT**, where collectively known as **mucosa-associated lymphoid tissues (MALT)** and these also may be considered as **lymphoid organs**.



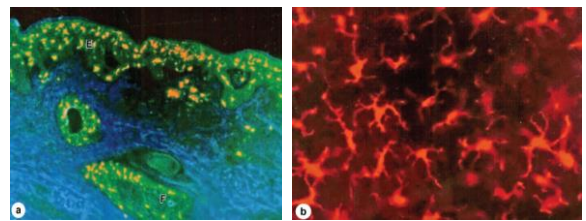
Distribution of Lymphoid Organs

Tip: ANALOGY

Human body	Nation/country
– Immune system	– Defense minister
– Lymphocytes and other immune cells	– Soldiers or police men or military force
– Primary Lymphoid organs	– Military colleges
– Secondary lymphoid organs	– Military camps
– Connective tissues and other tissues	– Battle fields
– Antigens and/or Pathogenic agents or environmental insults	– Invaders or enemies
– Antigen presenting cells	– Detectives
– Antibodies	– Bullets or weapons

Antigen-Presenting Cells (APCs)

- APCs are found in many tissues and constitute a heterogeneous population of cells that includes **dendritic cells**, macrophages, and B lymphocytes
- under certain circumstances, a variety of other cells including vascular endothelial cells, dermal fibroblasts, thyroid epithelial cells and brain astrocytes may process and present antigens to T lymphocytes.
- Dendritic cells (not to be confused with cells of nervous tissue) occur not only in the lymphoid organs, but are also abundant in epidermis and many mucosae, where they are called **Langerhans cells**.
- A common feature of APCs is the presence of class II MHC molecules on their surfaces. CD4⁺ T (helper) cells interact with complexes formed by peptides and class II MHC molecules on APCs.
- However, CD8⁺ T (cytotoxic) cells interact with peptides complexed with class I MHC molecules which can be presented by any nucleated cell.



Langerhans cells are dendritic antigen-presenting cells of the epidermis and other epithelia of body surfaces, where they comprise an important defense against pathogens and environmental insults. Like other APCs they develop in the bone marrow, move into the blood circulation, and finally migrate into stratified squamous epithelia where they can be difficult to identify in routinely stained sections. (a): Section of immunostained skin showing Langerhans cells (yellow) abundant in hair follicles (F), where many microorganisms live, and throughout the epidermis (E). Keratin of the epidermis and follicles is stained green. X40. Antibody against langerin/CD207 and counterstained with anti-keratin. (b): Face-on view of an epidermal sheet stained using the same antibody showing that Langerhans cells form a network among the other epidermal cells which is difficult for invading microorganisms to avoid. Having sampled the invaders' antigens, Langerhans cells emerge from the epidermis and travel to the nearest lymph node to elicit lymphocytes that can mount a collective immune response. X200

Cells of the Immune System

- function primarily to defend the organism by mounting **two types of immune responses**:
 1. **humoral immune responses**: by producing antibodies against foreign substances (antigens)
 2. **cell-mediated immune responses**: by direct action of cytotoxic cells against microorganisms, tumor cells, transplanted cells, and virus-infected cells.

LYMPHOCYTES

- constitute 20-30 of circulating leukocytes in peripheral blood.
- their wide distribution and the constant traffic through the **blood, lymph, loose CT & lymphoid organs** provide the body with an elaborate and efficient system of surveillance and defense.
- classified as **T, B, and natural killer (NK) cells [null cells]**.

LYMPHOCYTES

- Clones of **T & B lymphocytes**, each comprising a small number of identical cells, have the ability to recognize & respond to a single **antigenic determinant (epitope)**, or a small group of closely related epitopes.
- exposure to antigen & one or more cytokines induces activation of resting T & B cells, leading to their proliferation and differentiation into **effector cells**.

A. T LYMPHOCYTES

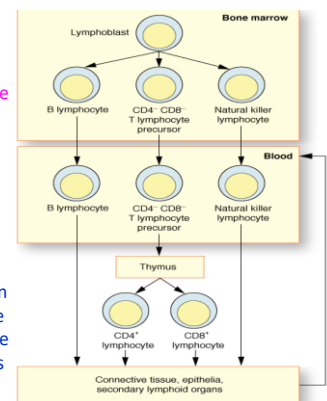
- constitute 65–75% of blood lymphocytes & include several functionally distinct subtypes.
- are responsible for cell-mediated immune responses and assist B cells in developing a humoral response to most antigens (called **thymic-dependent antigens**).
- recognize antigens that have been processed and displayed (presented) on the surface of APCs only.

1. Proliferation, Maturation and Differentiation of T Lymphocytes

- occurs in the thymus & involves the following events:
 - a. **Immunoincompetent lymphocyte precursors (lymphoblasts or prolymphocytes)**, derived from **lymphoid progenitors in the red bone marrow** migrate to the thymus, where they are termed **thymocytes**.
 - b. Within the **thymic cortex**, thymocytes undergo gene rearrangements [**preprocesses**] and begin to express antigen-specific T cell receptors (**TCR**), which are integral membrane proteins; i.e., the cells become **immunocompetent**.
 - c. The cortical thymocytes also undergo rapid mitosis and begin to express thymus-induced **CD markers (cluster of differentiation)** such as CD2, CD3, CD4, CD8 and CD25 on their surface.

Origin of the main types of lymphocytes

- B lymphocytes and natural killer (NK) lymphocytes are formed and become mature in the bone marrow and leave that compartment to seed the secondary lymphoid organs and transit through the blood to the epithelia and connective tissues.
- Immature CD4⁺ and CD8⁺ T lymphocyte precursors are transported by the circulation from the bone marrow to the thymus, where they complete their maturation and leave as either CD4⁺ or CD8⁺ cells.



- d. In the **thymic medulla**, some thymocytes lose CD4 and develop into CD8⁺ cells; others lose CD8 and develop into CD4⁺ cells.
- e. Medullary thymocytes also develop class I or class II HLA (human leukocyte antigen) molecules, the expression of the major histocompatibility genes.

2. T Lymphocytes Subtypes

a. T Helper (TH) Cells

- are CD4⁺ cells (normally 500-1500 cells/ μ liter of blood)
 - synthesize and release numerous growth factors, known as **cytokines (lymphokines)** specifically interleukin 2 (IL-2) & interleukin 4 (IL-4), following their activation.
1. The IL-2 & IL-4 induce B cells to respond to an antigenic stimulus.
 2. Other cytokines produced by TH cells modulate the immune response in diverse ways.

b. Cytotoxic T (TC) Cells

- are CD8⁺ cells
- after priming by an antigenic stimulus are induced by IL-2 to proliferate and differentiate into **effector cytotoxic T lymphocytes (CTLs)**, which modulate lysis of foreign cells and virus-infected self-cells via perforin.

c. T Suppressor (TS) (Regulatory) Cells

- are CD8⁺ and CD25⁺ cells and play crucial roles in switching off the immune response when the initiating stimulus is removed, maintaining unresponsiveness to self-antigens and suppressing excessive immune responses.
- produce **peripheral tolerance**, which backs up the central tolerance emerging in the thymus.
- can modulate the extent of the immune response by suppressing activity of TH cells, as well as TC cells.

d. T memory cells

- are long-lived, committed, immunocompetent cells that are formed during proliferation in response to an antigenic challenge.
- do not react against the antigen but remain in the circulation or in specific regions of the lymphoid organs.
- increase the size of the original clone and thereby provide a faster and greater secondary response (**anamnestic response**) against a future challenge by the same antigen.
- stimulated by TC cells

B. B LYMPHOCYTES

- originate and mature into immunocompetent cells within the liver during mid fetal life & within red bone marrow and/or other peripheral lymphoid organs during late fetal life and postnatally.
- have surface monomeric immunoglobulins IgM or IgD to recognize antigens; each B cell is covered by about 150,000 molecules of IgM.
 - all the immunoglobulin (Ig) molecules on a given B cell recognize & bind to the same antigenic determinant (epitope).
- The encounter of a B lymphocyte with the epitope it recognizes leads to several cycles of cell proliferation, followed by a redifferentiation of most of these lymphocytes into **plasma cells** and **B memory cells**.
- In most cases, the activation of B cells requires the assistance of **T helper cells**.

PLASMA CELLS

- are responsible for the humoral immune response.
- are terminal effector cells of immune system, which actively **synthesize** and **secrete** antibody against the same epitope as that recognized by the B cell from which it arose.
- **lack surface antibody/immunoglobulin.**
- Not all activated B cells, however, become plasma cells; some remain as long-lived **B memory cells**, which are able to react very rapidly to a second exposure to the same epitope.

C. NATURAL KILLER (NK) CELLS

- are one type of **null cell**, a **small group of peripheral blood lymphocytes** that **lack the surface determinants** characteristic of T and B lymphocytes.
- Their name derives from the fact that they attack virus-infected cells, transplanted cells, and cancer cells without previous stimulation; for this reason they are involved in what is called an **innate immune response**.
- exhibit an apparently **non-specific cytotoxicity** against tumor cells and virus-infected cells.
- also can kill **specific target cells** that have antibodies bound to their surface antigens in a process known as **antibody-dependent cell cytotoxicity (ADCC)**.
- **macrophages, neutrophils & eosinophils** also exhibit **ADCC**.

Major Differences Between T Cells and B Cells

Features	T cells	B cells
Directly involvement in the humoral antibody response	Yes; can facilitate or down regulate the response	Yes
Directly involvement in the cell-mediated immune response	Yes	No
Effector cell produced in immune response	Cytotoxic T cell= killer cell	Plasma cell
Identifying markers	T-cell receptors and CD4 and CD8	Surface immunoglobulins M/D

Approximate percentage of lymphocytes in lymphoid organs

Lymphoid Organs	T lymphocytes (%)	B lymphocyte (%)
Thymus	100	0
Red Bone marrow	10	90
Spleen	45	55
Lymph nodes	60	40
Blood	70	30

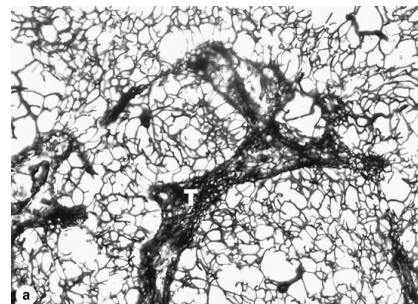
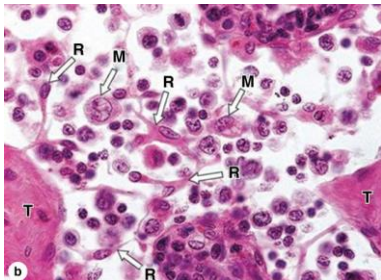
D. MACROPHAGES

- function both as antigen-presenting cells (APCs) and as **cytotoxic effector cells** in ADCC.
- produce interleukin 1 (IL-1), which helps **activate** TH cells and several other cytokines that influence the immune response or hematopoiesis.
- also secrete prostaglandin E2 (PGE2), which **decreases certain immune responses**.
- are activated by γ (gamma) **interferon**, which is also known as **macrophage-activating factor**.
- this activation increases the phagocytic and cytotoxic activity of macrophages.

Lymphoid Tissue

- is CT characterized by a rich supply of lymphocytes.
- exists freely diffusely within ordinary CTs OR surrounded by definite capsule, forming discrete lymphoid organs.
- Because lymphocytes have very little cytoplasm, lymphoid tissue filled with such cells stains dark blue in H&E stained sections; lymphoid tissues appear black mass.
- basically made up of free cells, typically with a rich network of reticular fibers (type III collagen) and reticular cells/ supporting cells.
- In most lymphoid organs, the reticular fibers are produced by a fibroblastic cell called a **reticular cell**, whose many processes rest on the fibers.
- The thymus is an exception in that its cells are supported by a reticulum of unusual epithelial reticular cells.

Cells of typical lymphoid tissue include the fibroblast-like reticular cells (R) which produce and maintain the trabeculae (T) and reticulin framework. Many cells are loosely attached to the reticulin fibers, including macrophages (M) and many lymphocytes. X240. H&E.



A three-dimensional framework of reticular fibers (collagen type III) supports the cells of most lymphoid tissues and organs (except in the thymus). Areas with larger spaces between the fibers offer more mobility to cells than areas in which the fiber mesh is tight, such as in trabeculae (T) and the cells here are generally more stationary. X10. Silver impregnation

LYMPHOID TISSUES

- Based on their organization, lymphoid tissues exist in two different morphological forms:

1. Diffuse lymphoid tissues or
2. Nodular lymphoid tissues.

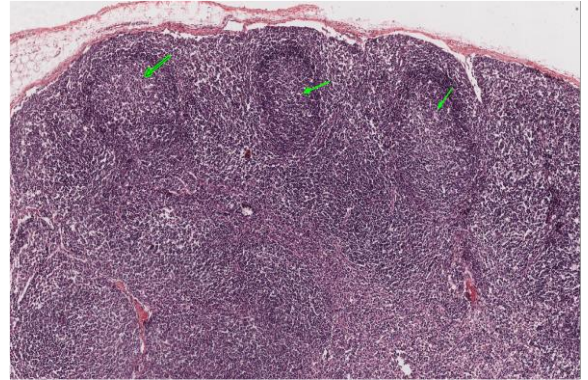
1. Diffuse lymphoid tissues or

- Scattered especially in any loose CT deep to the epithelia

2. Lymphatic nodules (lymph follicles)

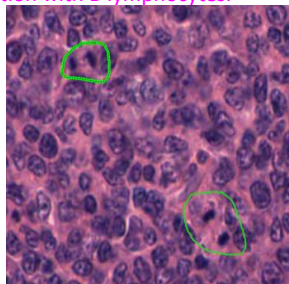
- In **nodular lymphoid tissue** groups of lymphocytes are arranged as spherical masses called **lymphatic nodules** or **lymph follicles**, containing primarily B lymphocytes.
- are transitory, dense, spherical accumulations of lymphocytes (mostly B cells)
- the dark peripheral region of nodules (**corona**) is composed largely of small, densely packed naïve B lymphocytes.

Lymphatic nodules (lymph follicles) in the outer cortex of a lymph node



Germinal Center

- In a germinal center, antigens bind surface proteins of FDCs in several ways (not including MHC proteins) but are not internalized or degraded by these cells.
- Rather antigen is retained by FDCs for extended periods (months to years) for interaction with B lymphocytes.
- As shown here, lymphoblasts undergo mitosis to amplify their colony after exposure to antigens.
- The marginal zone contains T lymphocytes which are mainly TH (CD4+) cells.



DISCRETE LYMPHOID ORGANS

1. THYMUS

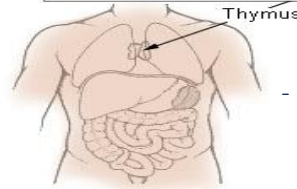
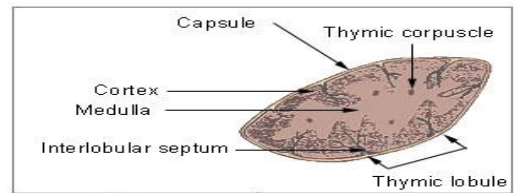
- a **lymphoepithelial, primary and central lymphoid organ** located in the mediastinum;
- attains its peak development during childhood.
- has a **dual** embryonic origin from **endoderm (3rd PP)** and **mesoderm**; nonthymic lymphoid organs however, are derived exclusively from mesenchyme (**mesoderm**).
- In addition to secreting thymic hormones, the adult thymus primes thymocytes (T-cell precursors) before releasing them to the periphery.
- Its lymphocytes arise in the bone marrow from lymphoid progenitors, cells of mesenchymal origin that invade an epithelial primordium that has developed from the **endoderm** of the paired third pharyngeal pouches (epithelial reticular cells).

- Thymus is surrounded by a connective tissue capsule that sends short septa into the parenchyma dividing it into **incomplete lobules**, each of which contains a darker, highly cellular outer **cortex** and a paler, less cellular central **medulla**.

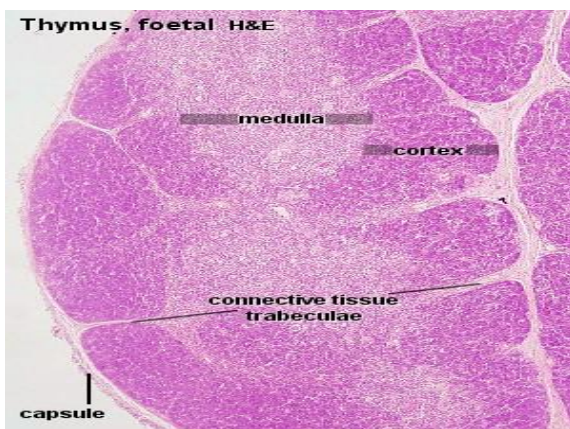
1. THYMIC CORTEX

- it is the outer region of the thymus composed of an extensive population of T lymphoblasts (also called **thymocytes**) and macrophages in a stroma of **epithelial reticular cells**.
- is supplied by arterioles **in the septa** which gives rise to capillary loops of **continuous type** that are surrounded by macrophages and pericytes.
- **is the region in which T-cell maturation occurs and it is immunologically protected region.**

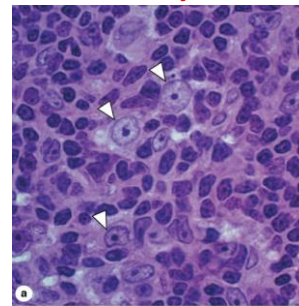
THYMUS



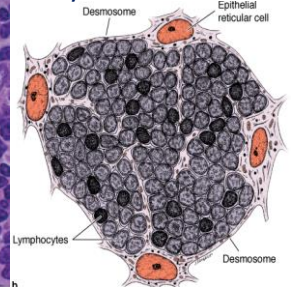
- Thymus lacks **lymphatic nodules (lymph follicles)**.



Cortex of Thymus



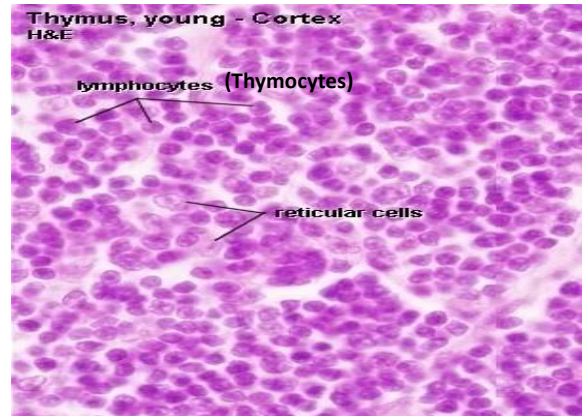
Notice absence **lymph follicles** in the thymus.



The cortical zone of an active thymus is packed with lymphoblasts that proliferate as well as undergo positive and negative selection in that region. The lymphoblasts are supported in a meshwork of epithelial reticular cells (arrowheads). X400. PT. (b): The epithelial reticular cells extend long processes bound together by desmosomes to make the framework for the lymphocytes. The epithelial reticular cells also secrete polypeptide factors that promote T cell maturation.

b. THYMOCYTES

- are immature T lymphocytes or T lymphocyte precursors present in large numbers within the thymic cortex in different stages of differentiation.
- are surrounded by processes of epithelial reticular cells, which help to segregate thymocytes from circulating antigens during their maturation.
- migrate toward the medulla as they mature, however most T cells die in the cortex and are phagocytosed by classical macrophages.



C. Blood-Thymus Barrier

- exists in the cortex only, making it an immunologically protected region.
- ensures that antigens escaping from the bloodstream do not reach developing T cells in the thymic cortex.
- consists of the following layers: continuous endothelium of blood capillaries and its complete basal lamina, perivascular connective tissue and cells (e.g. pericytes and macrophages), and epithelial reticular cells and their basal laminae.

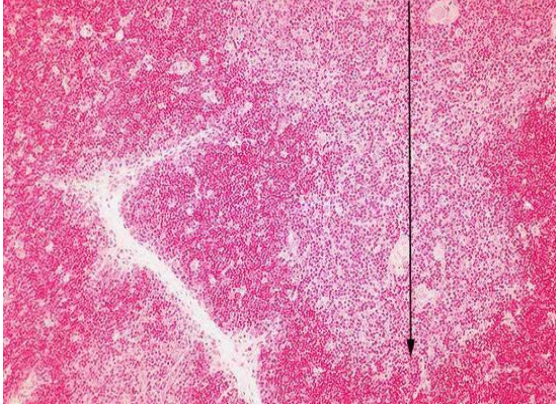
2. THYMIC MEDULLA

- is continuous between adjacent lobules.
- contains large numbers of epithelial reticular cells and mature T lymphocytes, which are loosely packed, causing the medulla to stain lighter than the cortex.

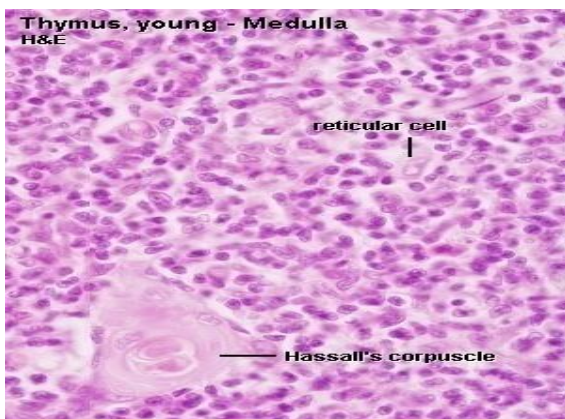
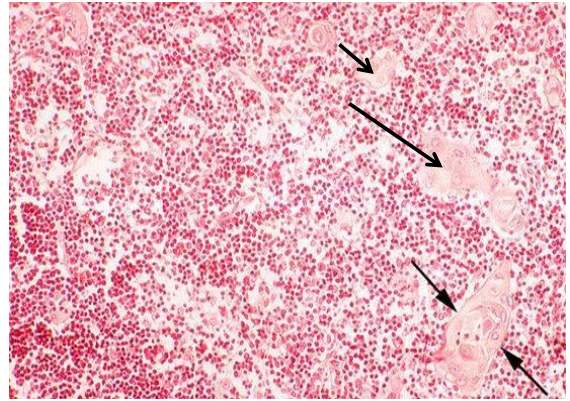
THYMIC MEDULLA

- is the region from which mature T cells exit the thymus via venules and efferent lymphatic vessels.
- the T cells then migrate to nonthymic lymphoid structures (peripheral lymphoid organs).
- also contains a cytotreticulum of epithelial reticular cells, many less densely packed differentiated T lymphocytes, and structures called thymic (Hassall's) corpuscles, which are characteristic of this region.
- Thymic corpuscles consist whorl [curl]-like accretions [growth] of degenerating epithelial reticular cells arranged concentrically, filled with keratin filaments, and sometimes calcified and increase in number with age.
- The function of Hassall's bodies has not been established.

Thymus medulla



Hassall's corpuscles (in the thymic medulla)

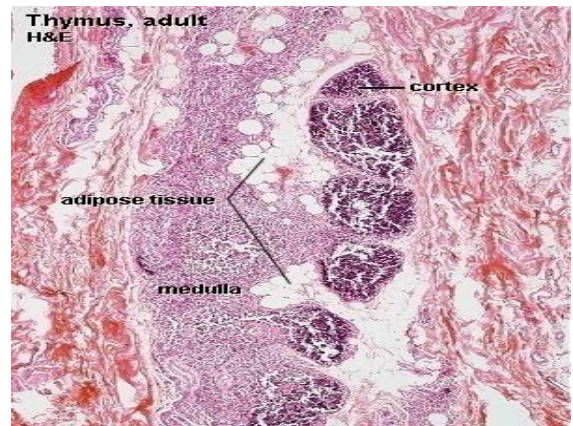


Involution of the Thymus

- slow non-hemorrhagic process which begins near the time of puberty,
 - involves two distinct histological processes, **fatty infiltration** and **lymphocytic depletion**, which occur independently on a different time scale.
 - Fatty infiltration and lymphocyte depletion results in disruption of cortico-medullary organization.
- 1. Fatty Infiltration**
- Adipocytes first begin to appear in the supportive tissue stroma around the blood vessels at birth, their number slowly rising until puberty when the rate of fatty infiltration increases markedly.
 - Consequently, fatty filtration occurs in the intercellular septa, spreading out into the cortex and later the medulla.

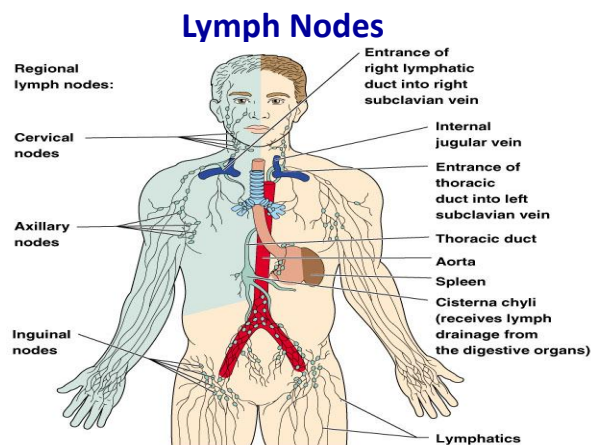
2. Lymphocyte Depletion

- Lymphocyte number begins to fall from about one year of age, the process continuing then after at a constant rate.
- Despite this, the thymus continues to provide a supply of mature T lymphocytes to the circulating pool and peripheral lymphoid organs and tissues.
- Lymphocyte depletion results in collapse of the epithelial frame work.
- However, cords of epithelial cells persist and continue to secrete thymic hormones through out life time.



2. LYMPH NODES

- are small, encapsulated, ovoid to kidney-shaped organs about 100,000 in an adult.
- are scattered along lymphatic vessels throughout the body, but are concentrated in the **neck, axilla, abdomen, and groin**.
- have a convex surface that receives afferent lymphatic vessels, and a concave surface (the hilum) where arterioles enter and efferent lymphatic vessels and venules exit.
- are surrounded by a **capsule of dense collagenous CT**, which sends trabeculae which extend for a variable distance into the substance of the node.
- are divided into **cortical** and **medullary** regions.
- possess a reticular CT stroma composed of a supportive framework rich in reticular fibers intimately associated with the capsule and trabeculae.



Lymph Node Structure

- Located along lymphatic vessels; most are kidney-shaped

CORTEX

Outer cortex

- Contains follicles (lymphatic nodules) – collections of egg-shaped aggregates of B lymphocytes.

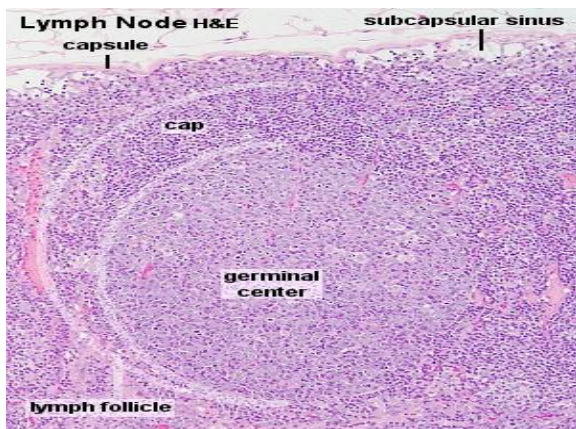
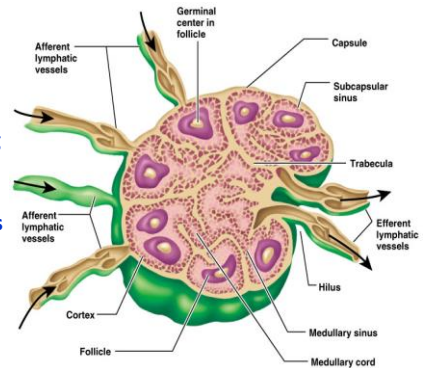
Inner cortex (Paracortex)

- Does not contain lymphatic nodules/follicles
- Contains mainly T cells and lymphoid dendritic cells that enter from other lymph node or from other tissues.

MEDULLA- Inner part

- Contains phagocytic cells, plasma cells, B cells, re-circulating lymphocytes and macrophages

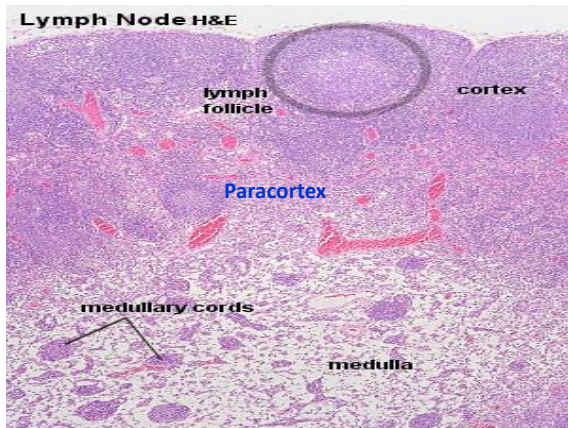
Lymph Node Structure



- A changing population of immune cells is suspended on reticular fibers throughout the cortex, paracortex, and medulla.

1. Cortex of Lymph Nodes

- lies deep to the capsule from which it is separated by a subcapsular sinus.
- **incompletely subdivided** into smaller intercommunicating compartments by CT trabeculae derived from the capsule.
- contains the following major components:
 1. many CT reticular cells, macrophages, APCs, and lymphocytes
 2. lymphatic nodules with or without germinal centers,
 3. Cortical sinuses, and
 4. the paracortex.

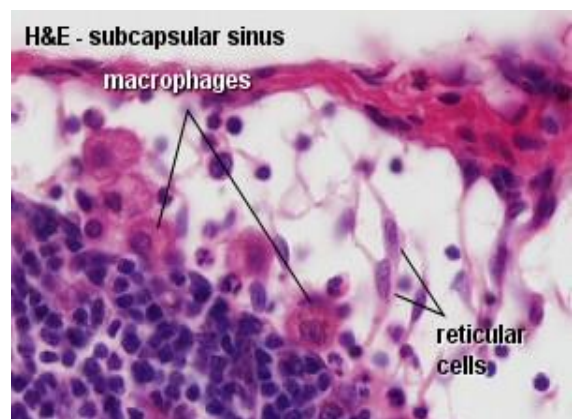


b. Lymphatic Sinuses

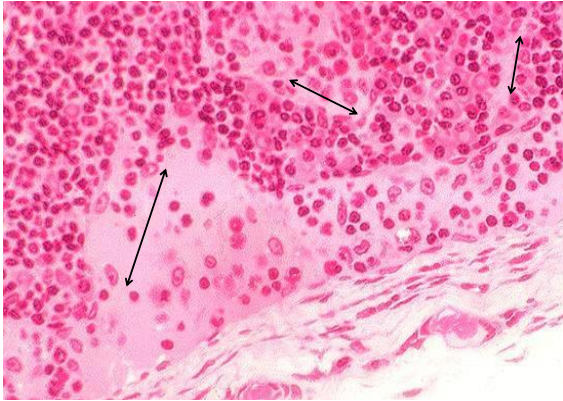
- are broad interconnected, endothelium-lined lymphatic channels that extend beneath the capsule (**subcapsular sinus**) along the trabeculae into the cortex (**trabecular/cortical sinus**) and then into the medulla (**medullary sinus**).
- Afferent lymphatic vessel divides into small multiple branches outside the node.
- These multiple lymphatics pierce the capsule and empty their content into lymph node at subcapsular sinus.
- lymph percolates through the node and exit at the hilum via a single efferent lymphatic vessel.
- blood supply is via small artery branching out into arterioles then capillary bed, then venules and finally vein on the way out of the node at the hilum.

Subcapsular Sinuses and Cortical Sinuses

- **Subcapsular sinuses** are areas immediately beneath the capsule where the lymphoid tissue has wide reticular fiber meshes and abundant macrophages.
- Lymph containing antigens, lymphocytes, and APCs circulates around the wide spaces of these sinuses after being delivered there by the afferent lymphatic vessels.
- **Cortical sinuses**, running between the lymphoid nodules, which arise from and share the structural features of the subcapsular sinuses.
- They communicate with the subcapsular sinuses through spaces similar to those present in the medulla.

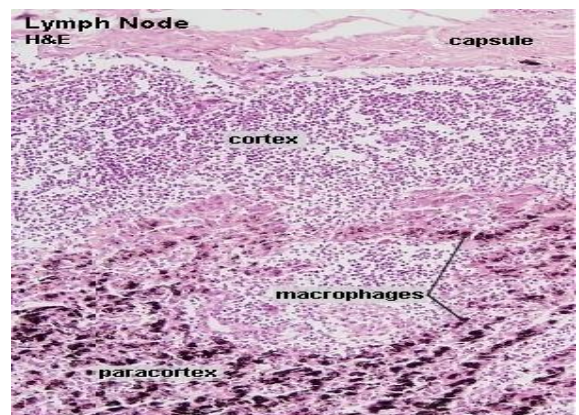
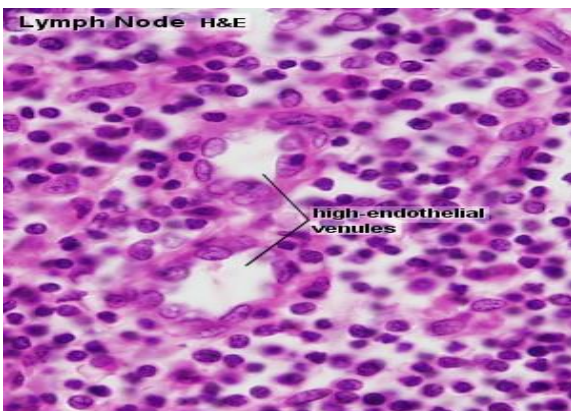


Cortical sinuses



c. PARACORTEX (INNER CORTEX)

- is located between the cortex (with nodules) and the medulla
- is composed of a **non-nodular** arrangement of mostly T lymphocytes (**the thymus-dependent area of the lymph node**).
- is the region where circulating lymphocytes gain access to lymph nodes via postcapillary (high endothelial) venules.
- **Post capillary venules** (high endothelial vessels recognized by their lining of cuboidal endothelium) are specific sites where lymphoid cells enter and exit blood vessels to and from lymph node, for or from **homing**.
- In a T cell dominated immunological response, the paracortical zone may extend into medulla, producing the so-called **paracortical reactions**.



- Activated T cells are then disseminated via the circulation to peripheral sites where much of their activity occurs.

2. MEDULLA OF LYMPH NODES

- lies deep to the paracortex and cortex **except at the region of the hilum**.
- is composed of medullary sinusoids and medullary cords.

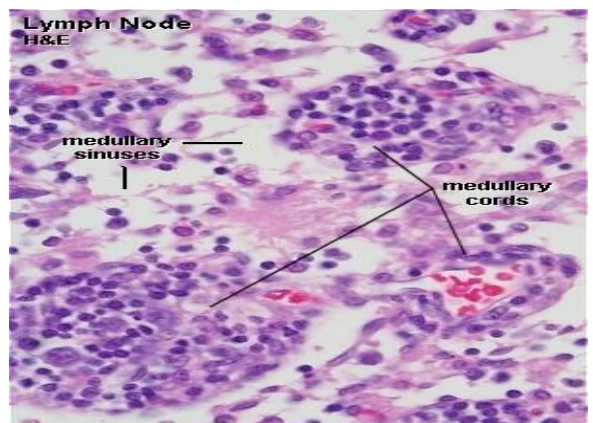
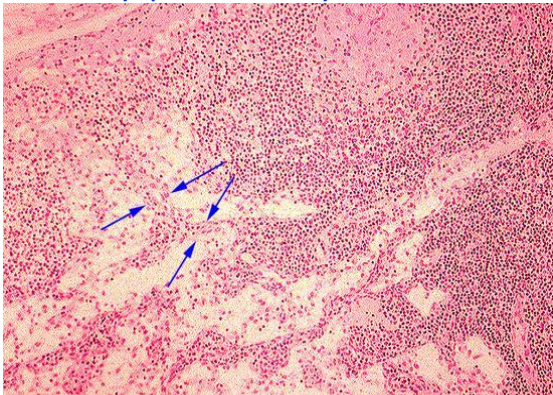
a. Medullary cords

- are branched cordlike extensions of lymphoid tissue arising from the paracortex.
- contain primarily lymphocytes, **plasma cells** and macrophages, many of which migrated from the cortex to enter the medullary sinusoids.
- are separated by dilated spaces, frequently bridged by reticular cells and fibers, called medullary sinuses.

b. Medullary sinuses

- are endothelium-lined spaces supported by reticular fibers and reticular cells, whose processes frequently span the sinusoids, retarding and causing turbulence in the lymph flow; they also contain microphages.
- receive lymph from the cortical sinuses.
- contain lymph, lymphocytes, often many macrophages, and sometimes even granulocytes if the lymph node is draining an infected region.
- these sinuses are continuous with the cortical sinuses and join at the hilum to deliver lymph to the efferent lymph vessel of the lymph node.

Lymph node medullary cords

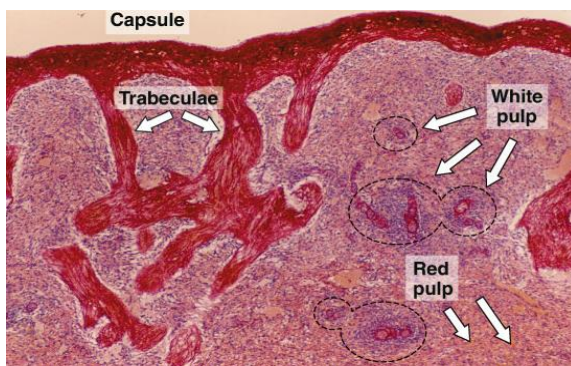


MEDICAL APPLICATION

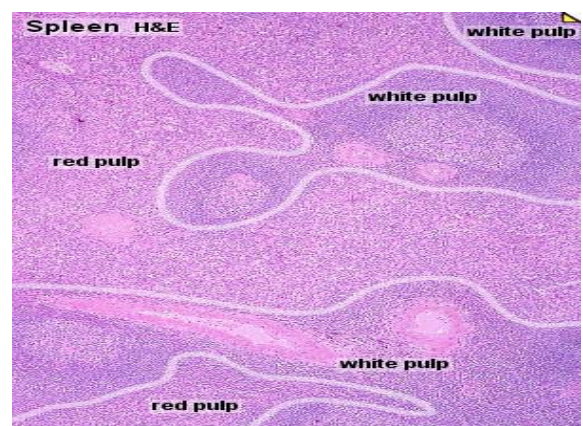
- As each satellite node receives lymph from a limited region of the body, malignant tumor cells that reach lymph node are distributed to other parts of the body via the efferent lymph vessels and blood vessels, a process known as **metastasis**.
- Infection and antigenic stimulation often cause lymph nodes to enlarge.
- These swollen nodules, which may be palpated under the skin as indicators of inflammation, have multiple germinal centers with active cell proliferation.
- Although plasma cells constitute only 1–3% of the cell population in resting nodes, their numbers increase greatly in stimulated lymph nodes.

3. SPLEEN

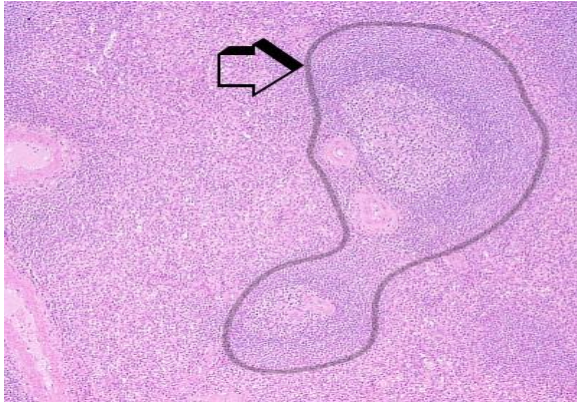
- is the largest lymphoid organ of the body.
- **acts as a blood stream filter.**
- is surrounded by a **capsule of dense regular** collagenous CT with interspersed with elastic fibers and smooth muscle cells. The capsule sends in **trabeculae**
- the capsule's internal aspect has a reticular CT stroma that extends along the trabeculae and forms the framework of the spleen.
- differs from the thymus and lymph nodes in that it lacks a cortex, medulla, and afferent lymphatics.
- is divided into regions designated as **red pulp** and **white pulp**, the latter [**white pulp**] contains **lymphoid nodules**.
- these two regions are separated by a **marginal zone**.



Section of spleen: the capsule is seen sending trabeculae to the interior of the organ. The red pulp occupies most of the microscopic field. Note the white pulp with its arterioles. Picrosirius stain.



White pulp of the spleen



Functions of Spleen

- filtration of particulates from blood (bacteria, Ag/Ig complexes, etc.)
- **storage of erythrocytes**
- phagocytosis of damaged or spent (aged) **erythrocytes** and **platelets**.
- proliferation of B and T lymphocytes
- production of antibodies by plasma cells.
- in other mammals, it is also a reservoir of blood
- **1. Vascularization of the spleen**
- is derived from the **splenic artery**, which enters the hilum and gives rise to trabecular arteries; these are distributed to the splenic pulps via trabeculae.

a. Trabecular arteries

- are derived from the **splenic artery** and found in the CT trabeculae
- enter the spleen parenchyma as arterioles enveloped by a sheath of T lymphocytes, the **periarteriolar lymphoid sheath (PALS)**, which is part of the white pulp.
- gives rise to these small branches which are ensheathed vessels designated **central arterioles**.

b. Central arterioles

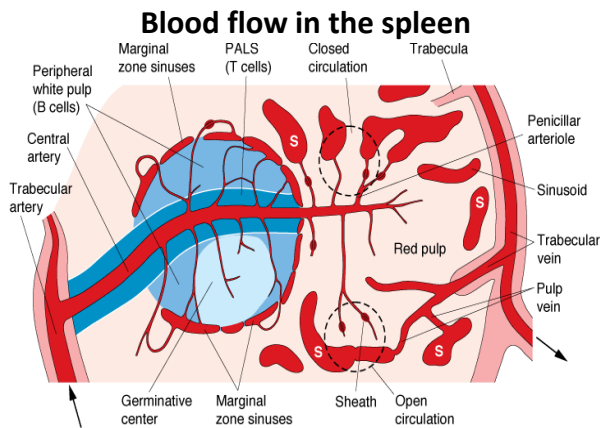
- supply the white pulp.
- gives rise to ensheathed capillaries branch but maintain lymphatic sheath until they leave the white pulp to form several straight **penicillar arterioles**.

c. Penicillar arterioles

- arise as straight branches of the central arterioles which arise almost at 90 degrees.
- enter the red pulp through the marginal zone and possess three regions: **pulp arterioles**, **macrophage-sheathed arterioles**, and **terminal arterial capillaries (ensheathed)**.
- the latter either drain directly into the splenic sinusoids (closed circulation) or terminate as open-ended vessels within the red pulp (open circulation).

d. Splenic venous sinusoids: found in the red pulp

- are drained by pulp veins, which are tributaries of the trabecular veins (lack muscular wall); in turn, drain into the splenic vein, which exit the spleen at the hilum.
- **Splenic artery [within the hilum]---trabecular artery [within the trabecula]-- central arteriole [within white pulp]-- penicillar arterioles [within red pulp]-- Terminal capillaries**



2. WHITE PULP OF THE SPLEEN

- includes **all** of the organ's lymphoid tissue (diffuse and nodular) such as lymphatic (splenic) nodules (mostly B lymphocytes) and PALS (mostly T cells) around the central arterioles.
- also contains plasma cells, macrophages and other APCs.

3. Marginal zone of the spleen

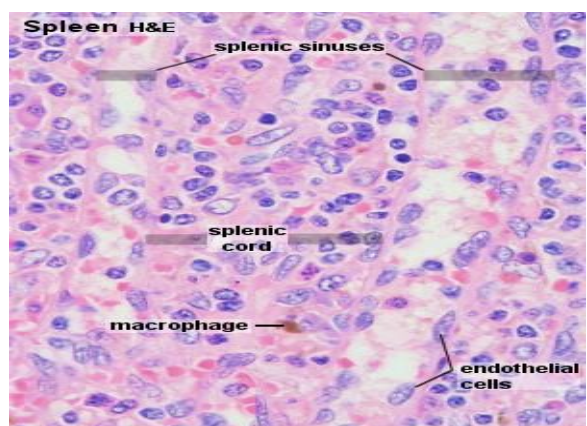
- is a **sinusoidal region** between the red and white pulp
- located at the **periphery of the PALS**.
- receives blood from the capillaries derived from the central arteriole and thus is the **first site where blood contacts the splenic parenchyma**.
- is richly supplied by avidly phagocytic macrophages and other APCs, and lymphocytes.
- is the region where T & B cells enter the spleen before becoming segregated to their specific locations.

4. RED PULP OF THE SPLEEN

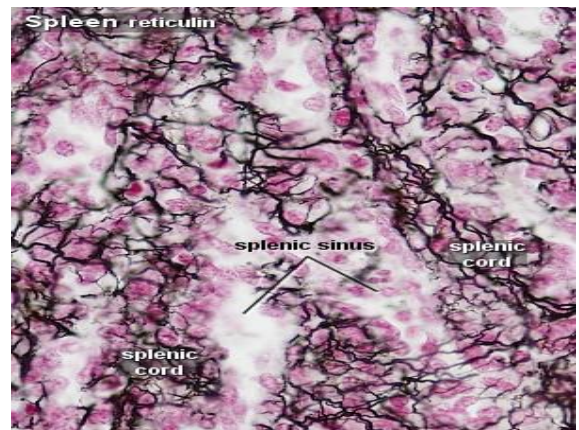
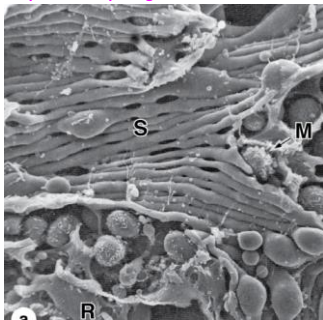
- is composed of an interconnected network of splenic sinusoids supported by a loose type of reticular tissue (**splenic cords**).
- is also supplied by avidly phagocytic macrophages and other APCs.
- **The macrophages** contain a pigment material, **hemosiderin** in their cytoplasm, which is a by product of degradation of spent and defective RBCs.

a. Splenic Sinusoids

- are lined by long, elongated, fusiform endothelial cells called **stave cells**, separated by relatively large intercellular spaces.
 - These cells are oriented in parallel with the sinusoid's blood flow.
- have a **discontinuous basal lamina**, underlying the fenestrated endothelium, and circumferentially arranged "ribs" of reticular fibrils.



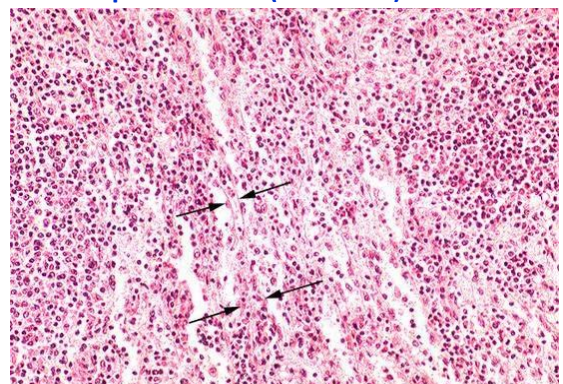
SEM clearly shows the parallel alignment of the stave cells (S), as well as many macrophages (M) in the surrounding red pulp (R). X500. (b) [NEXT SLIDE]: Silver-stained sections of spleen show black reticular fibers surrounding the sinuses, oriented 90 degrees to the long axis of the sinuses (S). These fibers appear similar to those in the surrounding splenic cords (C). The basement membrane of the stave cells is incomplete and open for the passage of cells. X400. Silver



b. Splenic cords [of Bilroth]

- contain plasma cells, reticular cells, and various types of immune cells (including macrophages, non-activated lymphocytes) enmeshed within the spaces of the reticular fiber network.
- processes of the macrophages enter the lumina of the sinusoids through the large intercellular spaces between the endothelial cells.

Splenic Cords (of Bilroth)

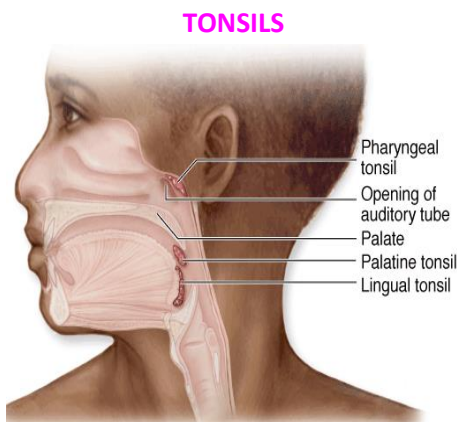


4. MUCOSA-ASSOCIATED LYMPHOID TISSUES (MALT)

- one of the largest lymphoid organs, containing up to 70% of all the body's immune cells.
- prominent in the loose CT deep to the epithelia linings of body cavities and tubes.
- Includes tonsils, Peyer's patch, GALT, BALT and **vermiform appendix**.
 - **GALT (Gut-associated lymphatic tissues)**
 - In the wall of esophagus, stomach, duodenum, cecum, colon and rectum)
 - **BALT (Breathing-associated lymphatic tissues)**
 - In the wall of larynx, trachea and bronchi
- can be organized as scattered clusters of lymphoid cells or as lymphatic nodules (lymph follicles).

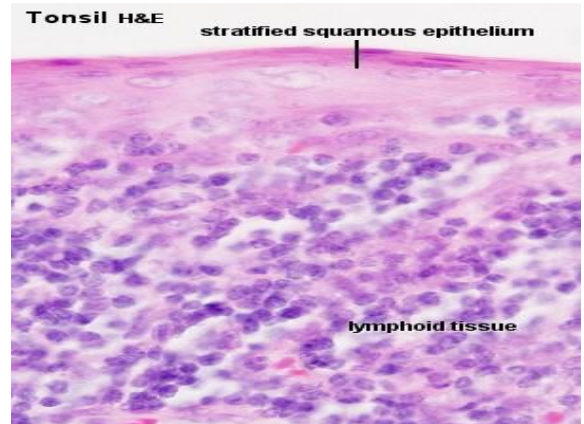
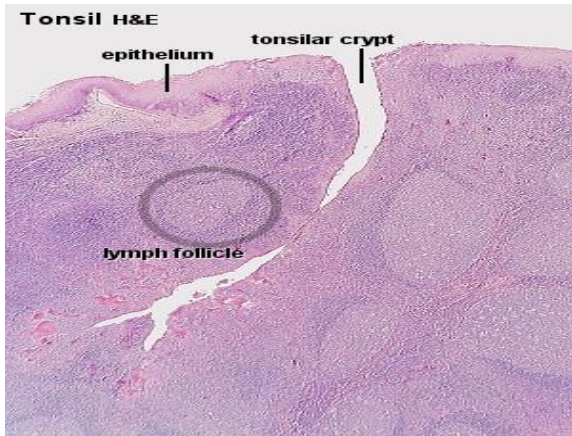
TONSILS

- Are large aggregates of lymphoid tissues around the oral cavity and pharynx (palatine, lingual, pharyngeal & tubal).
- are all located in the upper section of the digestive tract, lying beneath, but in contact with the surface lining epithelium.
- All these epithelia covering tonsils contain intraepithelial lymphocytes and dendritic cells.
- Except the lingual tonsils, tonsils are partially covered by CT capsule from inside only (hemicapsule).
- Lingual tonsil is non-capsulated.
- assist in combating antigens entering via the nasal, tubal and oral epithelia, forming **Waldeyer's ring** around these tubes.
- are members of MALT



1. PALATINE TONSILS

- are **two tonsillar** structures located in the posterior wall of oropharynx situated between the palatopharyngeal and palatoglossal arches.
- possess 10–20 **blind-ended deep** epithelial invaginations that penetrate the tonsil deeply, forming **crypts**, lined by **non-keratinized stratified squamous epithelium** which also covers the tonsils.
- **the crypts frequently contains cellular debris.**
- possess **lymphatic nodules (lymph follicles)**, some with germinal centers (secondary nodules).
- The area between the lymphoid follicles contain numerous T lymphocytes (and thus considered as thymus-dependent area of tonsils) and high endothelial venules.
- are separated from subjacent structures by a connective tissue capsule (**hemicapsule**).

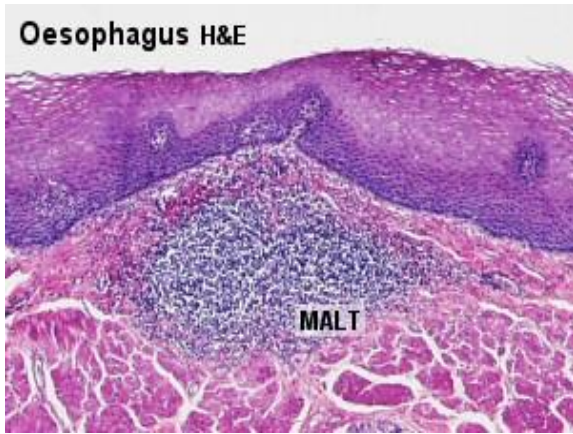


2. PHARYNGEAL TONSILS

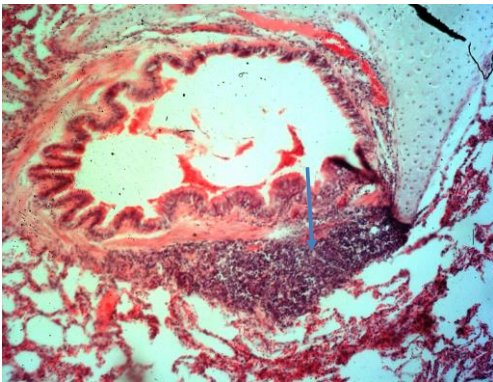
- located in the roof or upper part of the posterior wall of the nasopharynx near the entrance of Eustachian tube.
- covered by a **ciliated pseudostratified columnar epithelium (respiratory epithelium) of nasopharynx**
- contains several folds.
- seromucous secretions enter the folds from glands located in the CT deep to the tonsillar capsule.

3. LINGUAL TONSILS

- located on the dorsum of the posterior third (root) of the tongue.
- covered by **non-keratinized stratified squamous epithelium**.
- possess short **crypts**, which frequently contain cellular debris
- ducts of mucous glands often open into the base of these crypts.
- **are small and more numerous than the palatine and pharyngeal tonsils.**

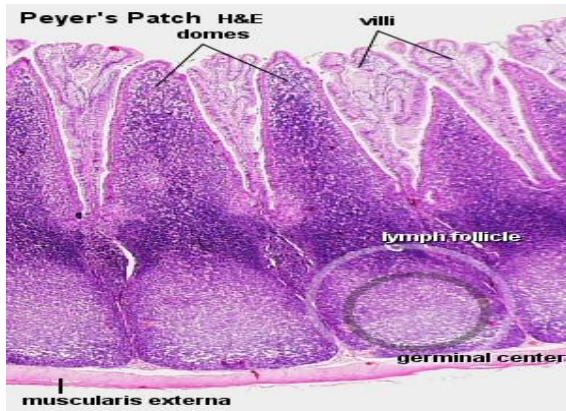


BALT in the wall of intrapulmonary bronchus



PEYER'S PATCHES

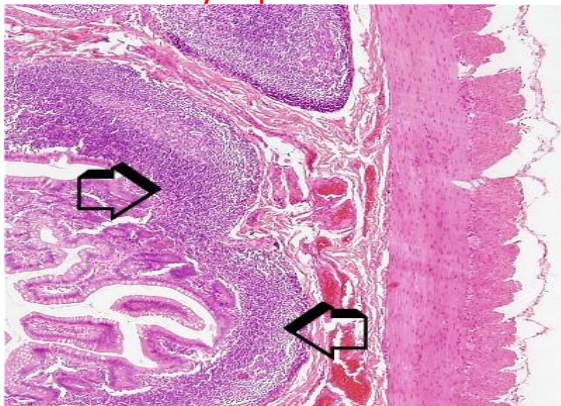
- are large aggregations of lymphoid tissues found in the mucosa and/or submucosa of ileum.
- are components of the gut-associated lymphoid tissues (GALT)
- they display prominent domes which bulge into the lumen.
- they may display nodular organizations
- the surface of the domes is covered by follicle-associated epithelium (FAE) which is simple cuboidal containing specialized antigen presenting cells called M (microfold) cells, with apical microfolds rather than the brush border and glycocalyx typical of enterocytes.
- M cells continuously sample antigens and microorganisms in the intestinal lumen.



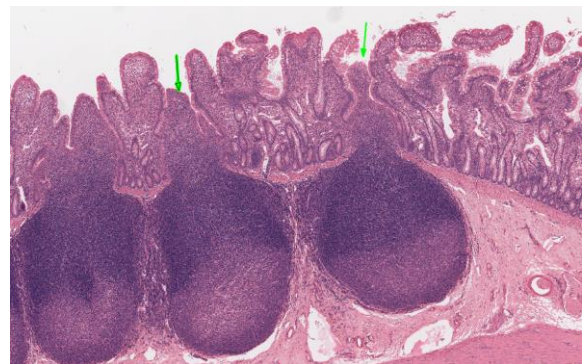
M Cells

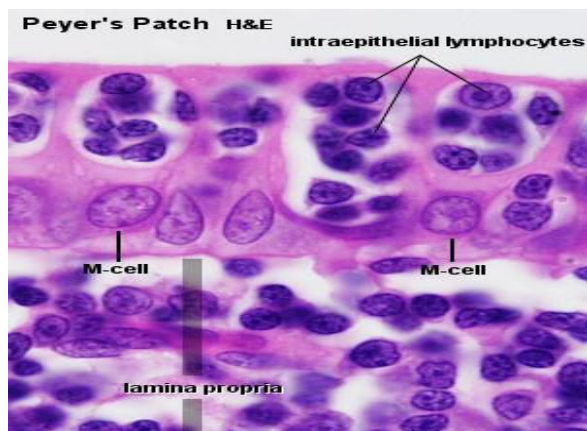
- Are epithelial cells characterized by a large basal intraepithelial pocket open to the underlying lymphoid tissue through a porous basement membrane.
- Antigenic material bound to the apical surface of M cells is rapidly translocated via transcytosis from the intestinal lumen to their intraepithelial pockets containing lymphoid dendritic cells, helper T cells, and B cells.
- Dendritic cells then take up the antigen, process it, and present it to lymphocytes and induce adaptive immune responses.
- T helper cells and B cells derived from these lymphocytes move away from the FAE and initiate adaptive responses to the antigens.
- These B cells give rise to plasma cells secreting IgA, which is transported by enterocytes into the intestinal lumen to bind and neutralize potentially harmful antigens.

Peyer's patch in ileum



Peyer's patch in ileum





Distinguishing Features of Lymphoid Organs

Features	Thymus	Lymph Node	Spleen	Tonsils
Role	primary	secondary	secondary	secondary
CT capsule with septa or trabeculae	present	present	present	Hemicapsule or absent
cortex & medulla	present	present	absent	absent
Lymphatics	Efferent only	both afferent & efferent	efferent only	efferent only
Main derivatives	T cells; thymic hormones	immunoglobulins; killer cells	immunoglobulins	immunoglobulins; killer cells
Exposure to foreign antigens	Blood, cortex is shielded	in lymph	in blood	in blood
Lymphoid nodules	Absent	present in cortex	present in white pulp	present
Plasma cells	Absent	present in GC and medulla	present in red pulp & in GCs	present in GCs
Recirculating lymphocytes	present in medulla only	present	present	present

CLINICAL CONSIDERATIONS

A. X-linked hypogammaglobulinemia

- a recessive, X-linked immunodeficiency affecting male newborns.
- characterized by a depressed (or absent) B-cell population and decreased levels of immunoglobulins, resulting in **impaired humoral immunity**.
- does not involve T cells; thus cell-mediated immune responses are normal.
- may be associated with the absence of tonsils and germinal centers in the lymphatic nodules.
- results in extreme susceptibility to infection with encapsulated bacteria (streptococcus, pneumococcus, etc).

B. DiGeorge's syndrome

- is also called **congenital thymic aplasia**.
- is characterized by the congenital absence of the thymus and parathyroid glands resulting from abnormal development of the third pharyngeal pouches.
- the length of the philtrum & clefts of the nose are reduced.
- is associated with abnormal cell-mediated immunity but relatively normal humoral immunity.
- usually results in death from tetany or uncontrolled infection.

C. Acquired Immunodeficiency Syndrome (AIDS)

- is caused by infection with **human immunodeficiency virus (HIV)**, which preferentially invades TH (CD4+) cells, causing a severe depletion in their number and thus suppressing both cell-mediated and humoral immune responses.

- is characterized by secondary infections with opportunistic microorganisms that cause pneumonia, toxoplasmosis, candidiasis, and other diseases.
- is also characterized by development of certain malignancies such as Kaposi's sarcoma and non-Hodgkin's lymphoma.

D. Hyperactive malarious splenomegaly

- is characterized by an enlarged spleen and very high serum levels of IgM and malaria antibody, as well as liver sinusoidal lymphocytosis.
- is likely due to recurring malaria, which disturbs T-cell control of the humoral immune response.

E. Hodgkin's disease

- is a malignancy involving neoplastic transformation of lymphocytes (most probably).
- occurs mostly in young adults and affects twice as many men as women.
- is characterized by the presence of Reed-Sternberg cells, which are giant cells with two large, vacuolated nuclei, each with a dense nucleolus.
- is marked by the painless, progressive enlargement of the lymph nodes, spleen, and liver.
- other symptoms include anemia, fever, weakness, anorexia, and weight loss.